CS481 HW3

Tomas Salaz

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Q1

I started coding on the CS Mahcines, specifically *moons* then on my local machine once I found out the ”TestQueueGuard” didn't hang locally. Running `'nproc'` on moons returned 16, and `'sysctl -n hw.physicalcpu'` on local (Mac) returned 10

Q2

In my commit messages I mentioned the moons broke so I did these tests on Mac. In my repo there will also be a file called `"results.txt"` with the dump from the terminal
The Yield Lock is the most Scalable. When comparing to all the other locks, the Yield Lock is consistently <6m for thread counts ≥64 where as the other locks seem to be 6ms ± .1ms
You Could argue that the Spin Lock scales better since it is faster at the 256 thread count, but Id rather opt for the consistently faster lock.

Q3

A. The Strong Scaling Limit based on the Results are:

Mutex: 32 Threads

Spin: 32 Threads

Yield: 128 Threads

Queue: 128 Threads

Semaphore: 128 Threads

B. The Point before the scaling limit where reduction happens in speedup is between 8-16 threads.
This Corresponds with the number of Cores on my Mac (10). Once the thread counts is bigger than 10, we have to start to pay the overhead for Context Switches which leads to little to no improvement in the Speed of the Locks.

Q4

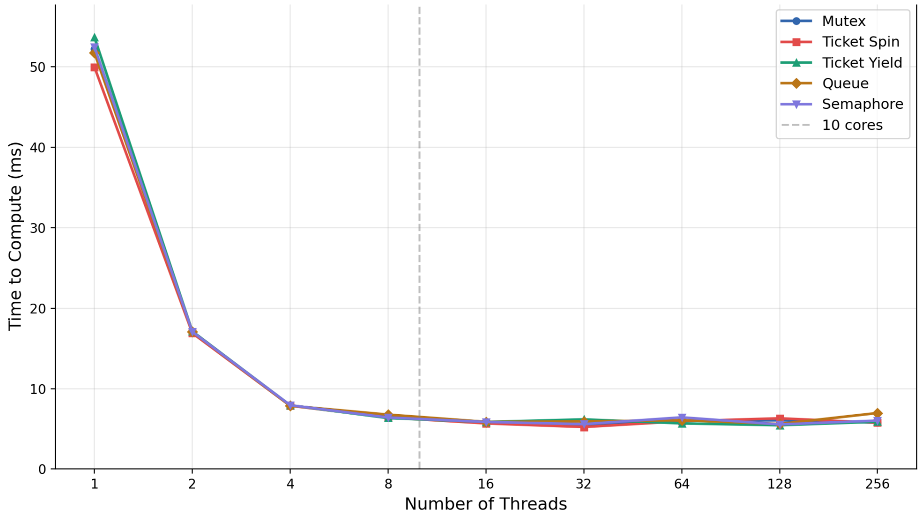

Figure 1: Enter Caption